Clinical Risk Assessment Report: Patient 77

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.42x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.1%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.42 [Avg (29th %ile)] (-2.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.14 [Avg (43th %ile)] (-1.0%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.4 [Top 7%] (-0.9%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) **Value: 64.0 [Avg (50th %ile)] (-0.9%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.13 [Avg (39th %ile)] (-0.8%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.42 [Bottom <1%] (-0.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.6 [Bottom 23%] (-0.6%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.04 [Avg (43th %ile)] (-0.6%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)